

Diagram 1: CFAR Sliding Window

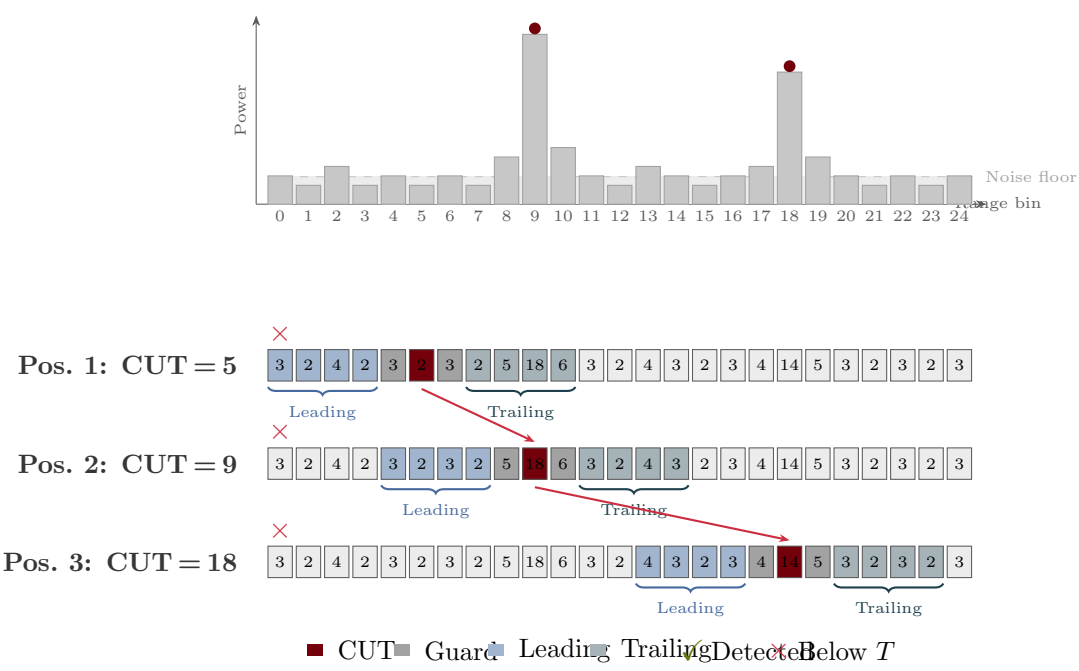


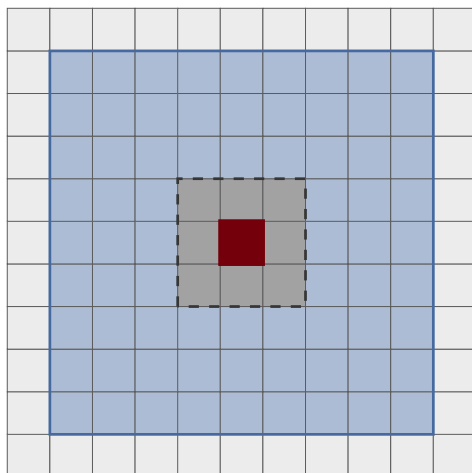
Diagram 1b: CFAR Variant Comparison

Raw Power

1	1	2	1	2	1	2	1	2	1	1
1	2	1	2	2	3	2	2	1	2	1
2	1	2	3	3	4	3	3	2	1	2
1	2	3	3	5	6	5	3	3	2	1
2	2	3	5	8	9	8	5	3	2	2
1	3	4	6	9	18	9	6	4	3	1
2	2	3	5	8	9	8	5	3	2	2
1	2	3	3	5	6	5	3	3	2	1
2	1	2	3	3	4	3	3	2	1	2
1	2	1	2	2	3	2	2	1	2	1
1	1	2	1	2	1	2	1	2	1	1



CA-CFAR



$$Z = \frac{1}{2N} \sum_{i=1}^{2N} x_i$$

CUT

Guard

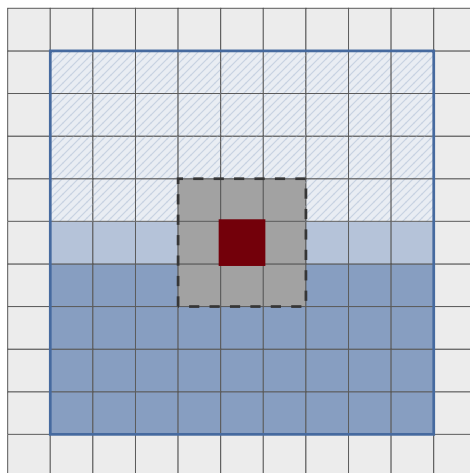
Selected Ref.

Raw Power

1	1	2	1	2	1	2	1	2	1	1
1	2	1	2	2	3	2	2	1	2	1
2	1	2	3	3	4	3	3	2	1	2
1	2	3	3	5	6	5	3	3	2	1
2	2	3	5	8	9	8	5	3	2	2
1	3	4	6	9	18	9	6	4	3	1
2	2	3	5	8	9	8	5	3	2	2
1	2	3	3	5	6	5	3	3	2	1
2	1	2	3	3	4	3	3	2	1	2
1	2	1	2	2	3	2	2	1	2	1
1	1	2	1	2	1	2	1	2	1	1



GO-CFAR



$$Z = \max(\bar{x}_L, \bar{x}_T)$$

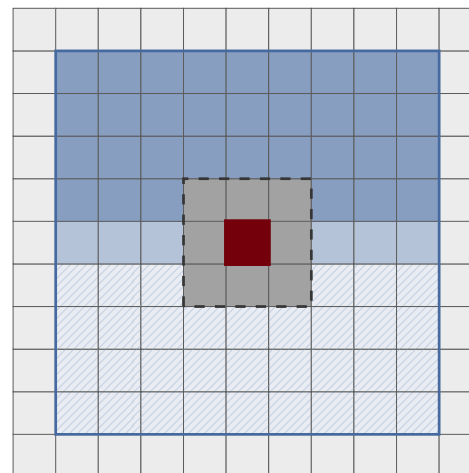
Unused Ref.

Raw Power

1	1	2	1	2	1	2	1	2	1	1
1	2	1	2	2	3	2	2	1	2	1
2	1	2	3	3	4	3	3	2	1	2
1	2	3	3	5	6	5	3	3	2	1
2	2	3	5	8	9	8	5	3	2	2
1	3	4	6	9	18	9	6	4	3	1
2	2	3	5	8	9	8	5	3	2	2
1	2	3	3	5	6	5	3	3	2	1
2	1	2	3	3	4	3	3	2	1	2
1	2	1	2	2	3	2	2	1	2	1
1	1	2	1	2	1	2	1	2	1	1



SO-CFAR



$$Z = \min(\bar{x}_L, \bar{x}_T)$$

Diagram 2: Range-Azimuth to PPI Mapping

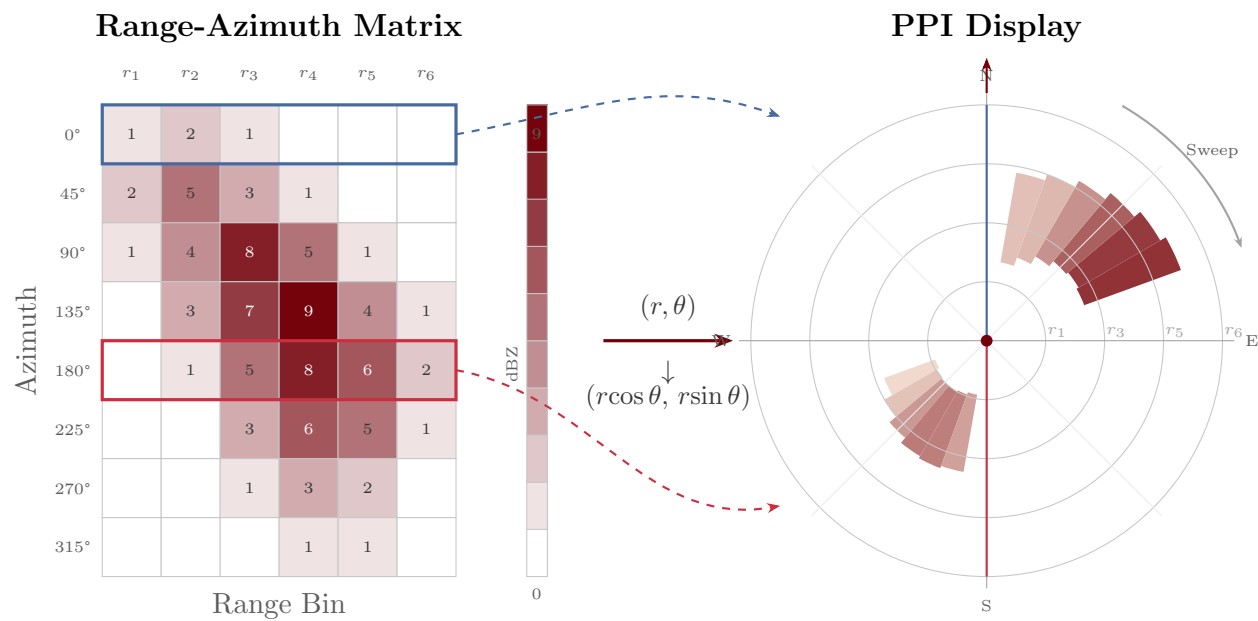


Diagram 3: Coverage Score Illustration

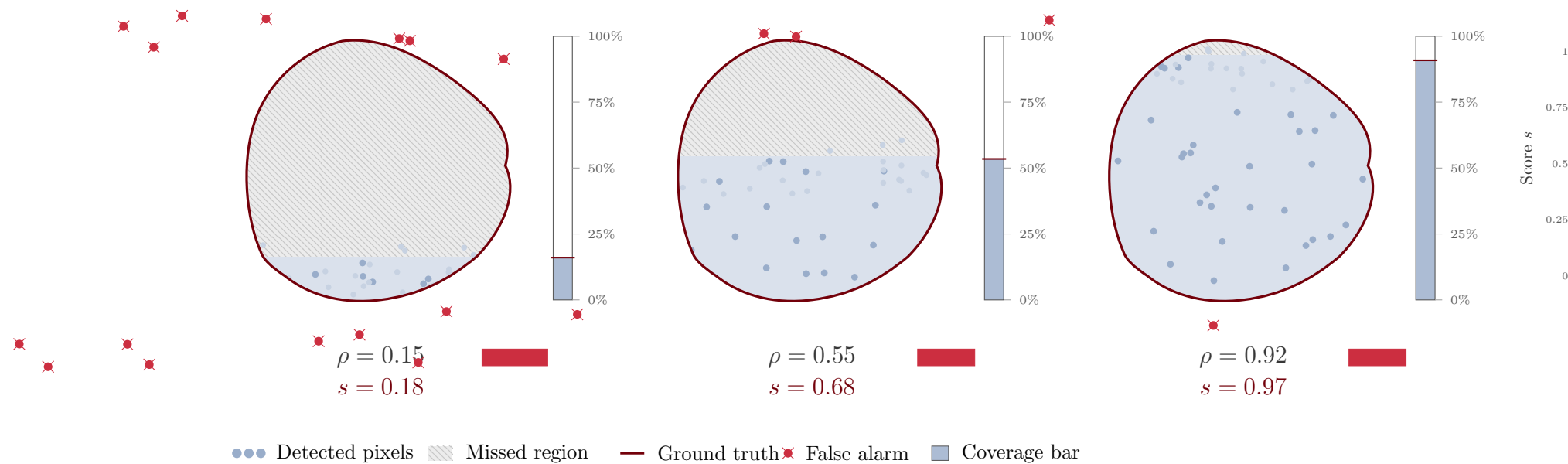


Diagram 4: Guard Cell Size Effect

(a) Guard Too Small

(b) Guard Appropriate

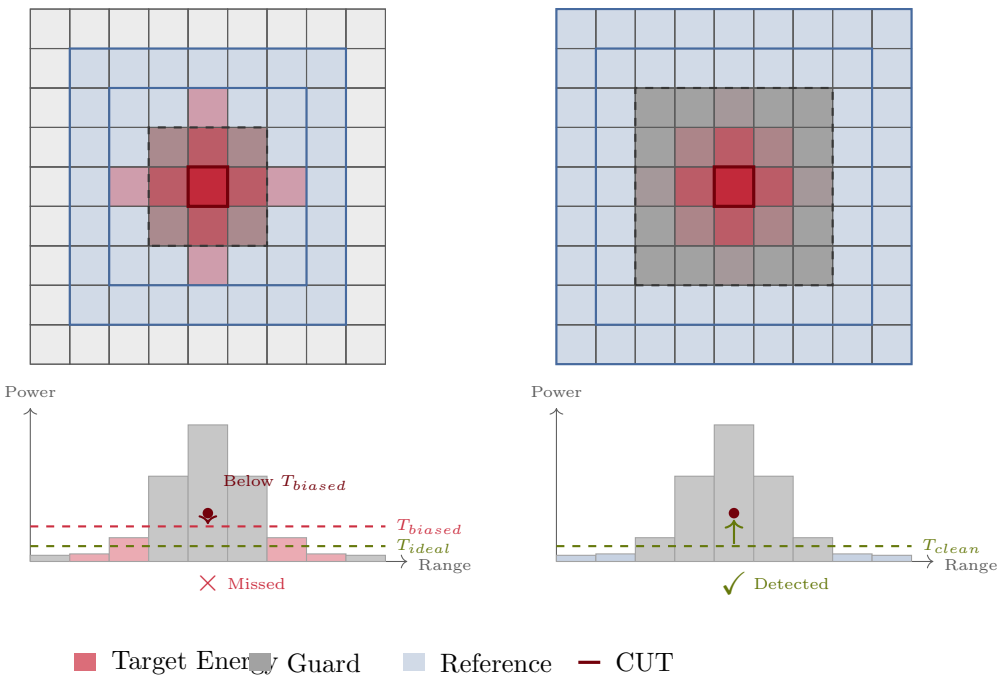


Diagram 5: Integral Image (Summed Area Table)

